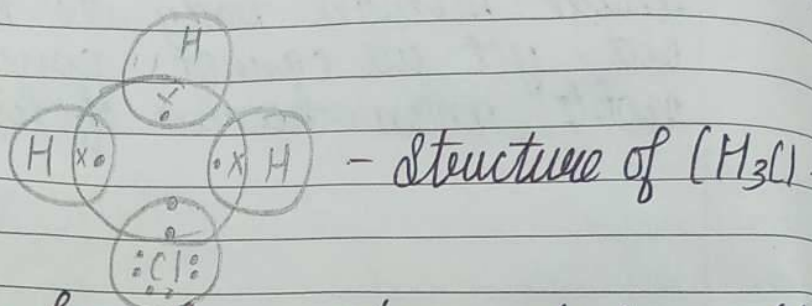


Chemistry

Ans 1. Fullerene, an allotrope of carbon which has 60 carbon atoms

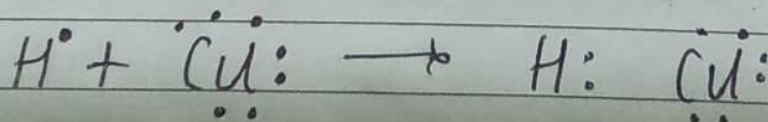
Ans 2.



Carbon has four valence electrons. It shares 1 electron each with 3 hydrogen atoms and 1 electron with chlorine. The bond between C and Cl atoms is covalent but due to higher value of electro-negativity of Cl, the C-Cl bond is polar in nature.

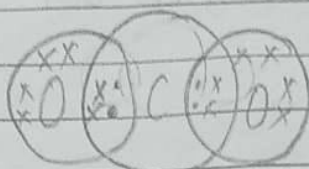
Ans 3.

(i) HCl



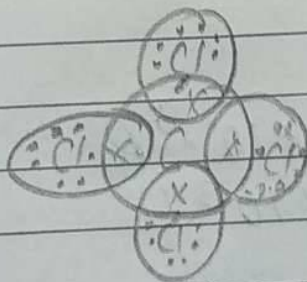
The Hydrogen in this gains one electron and the chlorine atom attains the octet of electron.

(ii) CO_2



Carbon has 4 valence electrons whereas oxygen has 6. So, one carbon atom forms 4 covalent bonds or 2 double bonds with two oxygen atom.

(iii) CCl_4



In this four chlorine atoms will combine with one carbon atom to form carbon tetrachloride which has a chemical formula as CCl_4 .